

Course Intro

Machine Learning Engineering

COMP3110

Week 1

- **1.1 Course intro:**
- **1.2 Law of the instrument**
- **1.3 Model selection, routing, and gateways**

1.1 COURSE INTRO

What is Machine Learning Engineering?

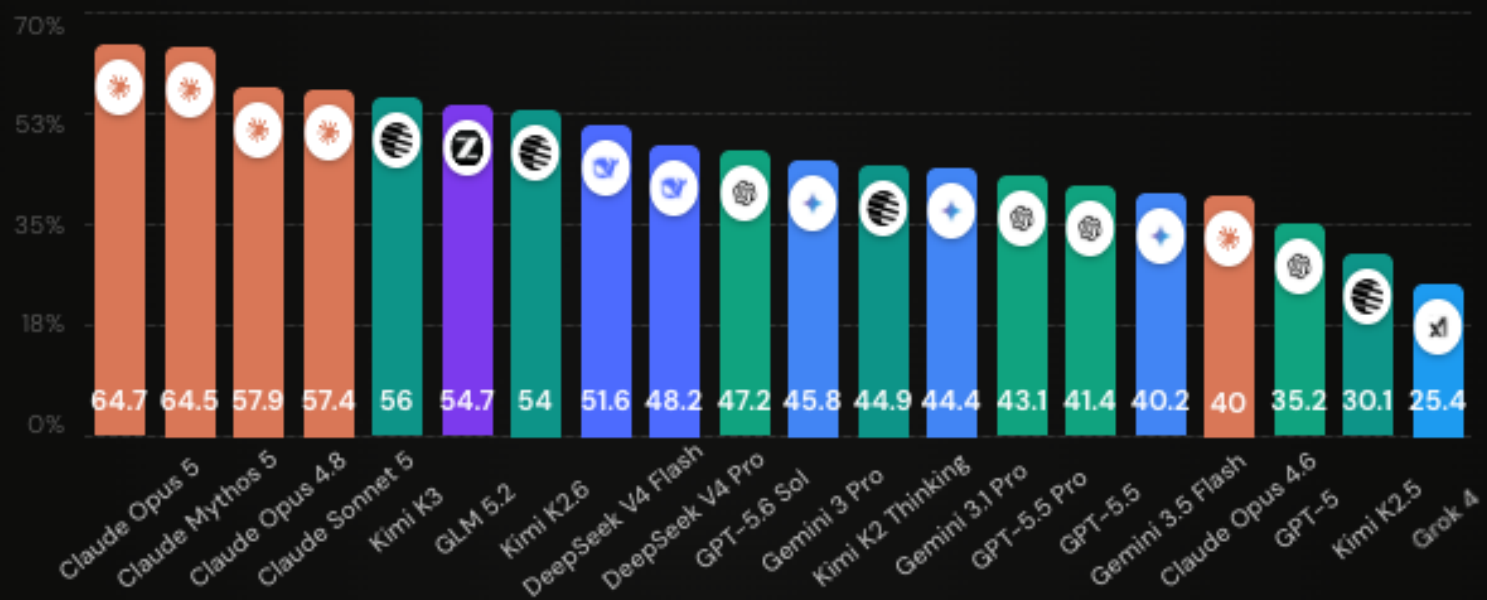
MLE vs ML

- **ML** is the algorithms: the maths and models. See courses like COMP9417.
- **MLE** is the engineer who ships ML into a working system. See the job ads below.
- **AI Engineer** also encompassed in 3110, basically LLMs in MLE.
- **COMP3110** is about building those ML/AI systems, end to end.

COMP3110 vs COMP9417

Topics	COMP3110	COMP9417
ML algorithms	No	Yes
Theoretical foundations	No	Yes
Datasets and problems provided	No	Yes
ML systems	Yes	Ish
Identifying problems	Yes	No
Measuring impact	Yes	No
Deployment	Yes	No

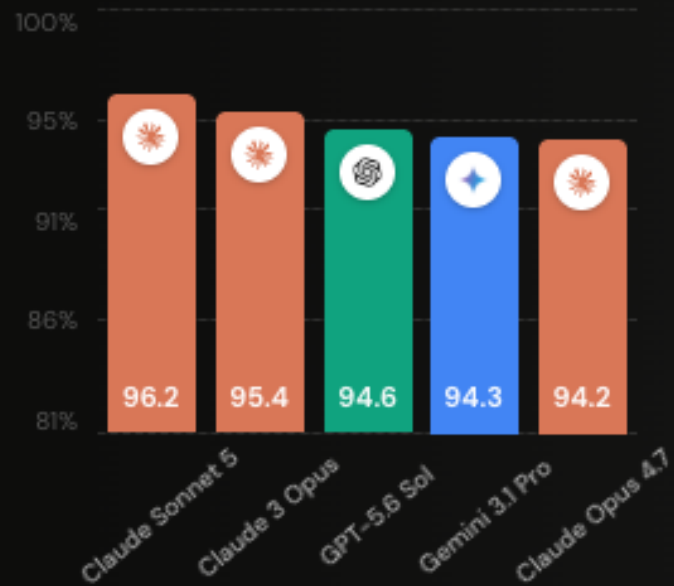
LLMS VS ML



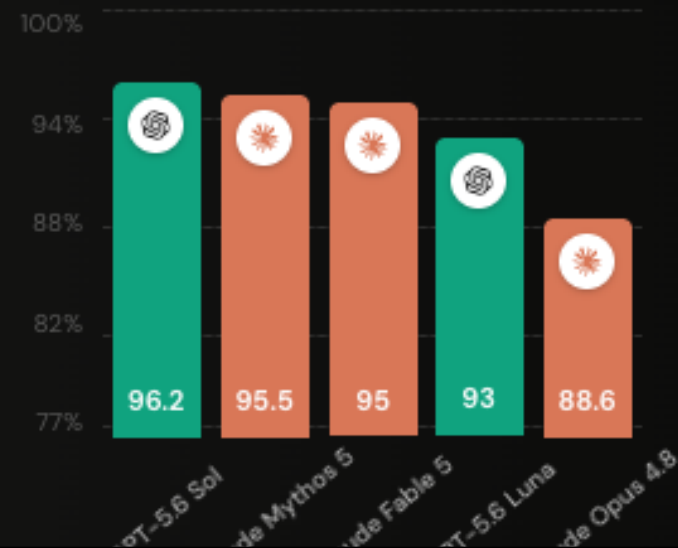
The frontier model just 1 year ago is today inferior to Qwen3.6-27b

Top models per tasks

Best in Reasoning (GPQA Diamond)



Best in Agentic Coding (SWE Bench)



COMP3110 is an accelerator



Accelerator

Started by you.
Accelerated with Google.

Google for Startups Accelerators serve top growth-stage startups with tailored technical, product, and leadership training from Google experts.

Find an Accelerator ↓



YC turns builders
into *formidable founders*^[1]

[1] “A formidable person is one who seems like they’ll get what they want, regardless of whatever obstacles are in the way.”

— Paul Graham

The term at a glance

- **Onboarding** (Weeks 1-2): form teams, pitch a problem, get started
- **Sprint 1** (Weeks 3-5): a prototype that runs end to end and ideally produces a metric.
- **Sprint 2** (Weeks 7-9): metrics, evals, and improving metrics.
- **Sprint 3** (Weeks 10): polish, Demo Day prep.
- **Demo Day** (Week 11): a short, live presentation of your deployed system.

What to expect

- An emphasis on the problem you choose to solve.
- A non-traditional lecture and content structure.
- Lots of support for you and your teams, including industry mentors.
- A lot of self-learning, so you can meet your own requirements.
- An in-person Demo Day to present your project.

What not to expect

- Highly specified tasks.
- Someone handing you the dataset, projects, metrics, or the solutions.

The course site

Everything (logistics, timetable, the accelerator, projects, mentors) lives here.



Mentors

Mentors are available throughout term to help with your MLE project.

Some guidelines:

- Mentors volunteer their time. Treat it with respect.
- Bring a specific problem that matches the mentor's skills.
- Be respectful, polite, grateful, and mindful of their time.

Assessment

- **Portfolio of Tasks (60%)**: evidence across onboarding and accelerator tasks, including individual contributions.
- **Final Project (40%)**: your final deliverable, including Demo Day performance and communication.

Guest lectures

- **Technical and non-technical:** A variety of topics from leaders in the field.
 - Let me know you have requests.
- **Scheduled on the Syllabus page.**
- **Attendance & Rules:** Most guest lectures will follow the Chatham House Rule. Important to respect this.¹

1. "When a meeting, or part thereof, is held under the Chatham House Rule, participants are free to use the information received, but neither the identity nor the affiliation of the speaker(s), nor that of any other participant, may be revealed." Chatham House.
<https://www.chathamhouse.org/about-us/chatham-house-rule> ↩

RECAP: COURSE INTRO

1. MLE is the engineering around a model: problem, impact, deployment.
2. The term runs as an accelerator: teams, sprints, Demo Day.
3. 60% portfolio of tasks, 40% final project.

Next, 1.2: before you reach for a model at all.